

ROCHAK BHATTARAI

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Research Profile

Cloud Engineer and Software Engineer with professional experience in Kubernetes, AWS, Docker, Terraform, Jenkins, and cloud-native application development. Passionate about intelligent cloud infrastructure, predictive autoscaling, distributed systems, and machine learning for cloud resource management. Seeking to pursue an MSc by Research to develop practical, scalable, and cost-efficient solutions for Kubernetes-managed cloud-native applications.

Research Interests

- Cloud Computing
- Cloud-Native Systems
- Kubernetes
- Distributed Systems
- DevOps Automation
- Infrastructure as Code
- Machine Learning for Cloud Resource Management
- Predictive Autoscaling
- Cloud Cost Optimisation

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering **2019 – 2023**

C.V. Raman Global University, Bhubaneswar, India

CGPA: **9.04 / 10**

- Awarded the prestigious **Study in India Scholarship**, a fully funded scholarship granted on academic merit by the Government of India.
- Relevant Coursework: Algorithms, Data Structures, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Cloud Computing, Artificial Intelligence.

Higher Secondary Education (+2) **2016 – 2018**

St. Lawrence Secondary School, Kathmandu, Nepal

Secondary Education (SLC) **2016**

Peace-Zone International Academy, Udayapur, Nepal

Professional Experience

Software Engineer **May 2024 – Present**

Aegis Software Pvt. Ltd., Kathmandu, Nepal

- Develop and maintain cloud-native enterprise applications using Kubernetes, Docker, C#, .NET, Jenkins and GitHub Actions.
- Design and implement CI/CD pipelines for automated build, testing and deployment.

- Deploy containerised applications on Kubernetes clusters and troubleshoot production workloads.
- Automate infrastructure provisioning using Terraform and Infrastructure-as-Code practices.
- Collaborate with cross-functional teams to improve deployment reliability, scalability and operational efficiency.
- Optimise application performance through container orchestration and deployment automation.

Cloud Engineer

July 2023 – April 2024

Aegis Software Pvt. Ltd., Kathmandu, Nepal

- Designed, deployed and managed cloud infrastructure using Amazon Web Services (AWS).
- Managed Kubernetes clusters (Minikube and Amazon EKS) for container orchestration and application deployment.
- Implemented Infrastructure as Code using Terraform for repeatable cloud provisioning.
- Automated software delivery pipelines using Jenkins, GitHub Actions and Git.
- Configured monitoring, logging and deployment automation for production cloud environments.
- Worked extensively with Docker, Kubernetes, Helm, Terraform and cloud-native technologies.

Leadership Experience

Chief Technology Officer (CTO)

Technology Innovation Nepal Pvt. Ltd.

- Lead the technical strategy and software architecture for cloud-native products.
- Coordinate software development activities and engineering teams.
- Design scalable application architectures and evaluate emerging technologies.
- Support research-driven software development and innovation initiatives.

Research Experience

Undergraduate Dissertation

360° Live Sports Streaming using Virtual Reality with Microservices and DevOps Practices

Duration: December 2021 – July 2023

- Designed and implemented a cloud-native microservices platform for hosting 360° live sports streaming services.
- Developed containerised microservices using Docker and orchestrated deployments using Kubernetes.
- Implemented Continuous Integration and Continuous Deployment (CI/CD) pipelines using Jenkins.
- Automated cloud infrastructure provisioning through Terraform.
- Investigated scalable deployment strategies, distributed application architecture, service orchestration and cloud-native software engineering.
- Evaluated system scalability, deployment automation, fault tolerance and operational efficiency using Kubernetes-based infrastructure.
- This research established the foundation for my proposed MSc research in intelligent Kubernetes resource management.

Proposed MSc Research

Machine Learning-Driven Predictive and Cost-Aware Autoscaling Framework for Kubernetes-Based Cloud-Native Applications

- Investigate the limitations of Kubernetes Horizontal Pod Autoscaler (HPA) in handling dynamic cloud workloads.
- Develop machine learning models capable of forecasting workload demand using historical Kubernetes metrics.

- Design a predictive autoscaling framework capable of proactively allocating resources before workload spikes occur.
- Integrate a cost-aware decision engine to optimise cloud infrastructure expenditure while maintaining application performance.
- Develop an intelligent descaling mechanism capable of safely reverting unnecessary scaling operations when predicted traffic does not materialise.
- Experimentally compare the proposed framework against Kubernetes' default Horizontal Pod Autoscaler using realistic cloud-native workloads deployed on Minikube and Amazon Elastic Kubernetes Service (EKS).

Research Interests

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|------------------------------|---|
| • Cloud Computing | • Infrastructure Automation |
| • Cloud-Native Systems | • Machine Learning |
| • Kubernetes | • Predictive Autoscaling |
| • Distributed Systems | • Cloud Resource Optimisation |
| • Microservices Architecture | • Cost-Aware Computing |
| • DevOps | • Intelligent Infrastructure Management |

Publications and Manuscripts

Building a Microservices-Based Application to Host Live 360° Sports Matches Using DevOps Practices

Status: Undergraduate research manuscript prepared based on the final year dissertation.

Current research interests focus on extending this work towards intelligent cloud resource management using machine learning-driven predictive autoscaling for Kubernetes-managed cloud-native applications.

Technical Skills

Cloud Platforms

Amazon Web Services (AWS): EC2, EKS, ECS, IAM, VPC, RDS, S3, CloudWatch, Lambda

Containerisation & Orchestration

Docker, Kubernetes, Helm, Minikube

Infrastructure as Code

Terraform, Ansible, CloudFormation

CI/CD & DevOps

Jenkins, GitHub Actions, GitLab CI/CD, Git, GitOps

Programming Languages

Python, Java, Go, C#, C++, Bash

Frameworks

.NET, ASP.NET, FastAPI, Django, React.js, Node.js

Databases

MySQL, MongoDB, PostgreSQL

Monitoring & Observability

Prometheus, Grafana, Kiali, Istio

Operating Systems

Linux (Ubuntu, CentOS), Windows Server

Awards & Academic Recognition

- Study in India Scholarship (Fully Funded), Government of India
- Star Performer Award – Excelerate (Project Management Internship)

Professional Certifications

- Google IT Automation with Python Professional Certificate
- Java 17 Masterclass
- Java Data Structures and Algorithms Masterclass
- Python for Software Engineering Bootcamp
- AWS Cloud and DevOps Professional Training

Languages

English (Professional Working Proficiency)

Nepali (Native)

Hindi (Professional Working Proficiency)

English Proficiency

IELTS - 7.5 not less than 6.5